

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403136
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Silcock; Alan et al.

Cannabinoid derivatives as pharmaceutically active compounds and methods of preparation thereof

Abstract

The present invention relates to a group of cannabinoid derivatives as pharmaceutically active compounds and methods of preparation thereof. The cannabinoid derivatives of the invention are analogues of cannabidiol (CBD). CBD is a non-psychoactive cannabinoid which has been used to treat various diseases and disorders. While such treatments hold promise, there remains a need in the art for more effective treatments and this has been brought about by way of the cannabinoid derivatives of the invention

Inventors:	Silcock; Alan (Cambridge, GB), Whalley; Benjamin (Cambridge, GB), Gray; Royston (Cambridge, GB), Straker; Hannah (Cambridge, GB), Tse; Karen (Cambridge, GB), Cobb; Alexander (London, GB), Bryant; Laura (Leigh-on-Sea, GB)
Applicant:	JAZZ PHARMACEUTICALS RESEARCH UK LIMITED (Sittingbourne, GB)
Family ID:	67839670
Assignee:	JAZZ PHARMACEUTICALS RESEARCH UK LIMITED (Sittingbourne, GB)
Appl. No.:	17/627946
Filed (or PCT Filed):	July 17, 2020
PCT No.:	PCT/GB2020/051723
PCT Pub. No.:	WO2021/014132
PCT Pub. Date:	January 28, 2021

Prior Publication Data

Foreign Application Priority Data

GB 1910389 Jul. 19, 2019

Publication Classification

Int. Cl.: A61K31/47 (20060101); A61K31/05 (20060101); A61K31/055 (20060101); A61K31/353 (20060101); A61K45/06 (20060101); A61P25/08 (20060101); C07C37/16 (20060101); C07C39/23 (20060101); C07C41/06 (20060101); C07D215/06 (20060101); C07D215/14 (20060101); C07D215/20 (20060101); C07D215/48 (20060101); C07D311/58 (20060101); C07D311/64 (20060101)

U.S. Cl.:

CPC A61K31/47 (20130101); A61K31/05 (20130101); A61K31/055 (20130101); A61K31/353 (20130101); A61K45/06 (20130101); A61P25/08 (20180101); C07C37/16 (20130101); C07C39/23 (20130101); C07C41/06 (20130101); C07D215/06 (20130101); C07D215/14 (20130101); C07D215/20 (20130101); C07D215/48 (20130101); C07D311/58 (20130101); C07D311/64 (20130101);

Field of Classification Search

CPC: A61K (31/47); A61P (25/08); C07C (37/16)

USPC: 514/312

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8632825	12/2013	Velasco Diez et al.	N/A	N/A
8790719	12/2013	Parolaro et al.	N/A	N/A
9017737	12/2014	Kikuchi et al.	N/A	N/A
9125859	12/2014	Whalley et al.	N/A	N/A
9474726	12/2015	Guy et al.	N/A	N/A
9675654	12/2016	Parolaro et al.	N/A	N/A
9949936	12/2017	Guy et al.	N/A	N/A
9949937	12/2017	Guy et al.	N/A	N/A
9956183	12/2017	Guy et al.	N/A	N/A
9956184	12/2017	Guy et al.	N/A	N/A
9956185	12/2017	Guy et al.	N/A	N/A
9956186	12/2017	Guy et al.	N/A	N/A
9962341	12/2017	Stott et al.	N/A	N/A
10039724	12/2017	Stott et al.	N/A	N/A
10092525	12/2017	Guy et al.	N/A	N/A
10098867	12/2017	Javid et al.	N/A	N/A

10111840	12/2017	Guy et al.	N/A	N/A
10137095	12/2017	Guy et al.	N/A	N/A
10220005	12/2018	Martinez-Orgado	N/A	N/A
10226433	12/2018	DiMarzo et al.	N/A	N/A
10583096	12/2019	Guy et al.	N/A	N/A
10603288	12/2019	Guy et al.	N/A	N/A
10653641	12/2019	Robson et al.	N/A	N/A
10709671	12/2019	Guy et al.	N/A	N/A
10709673	12/2019	Guy et al.	N/A	N/A
10709674	12/2019	Guy et al.	N/A	N/A
10729665	12/2019	Whalley et al.	N/A	N/A
10758514	12/2019	Liu et al.	N/A	N/A
10765643	12/2019	Guy et al.	N/A	N/A
10799467	12/2019	Whalley et al.	N/A	N/A
10807777	12/2019	Whittle	N/A	N/A
10849860	12/2019	Guy et al.	N/A	N/A
10918608	12/2020	Guy et al.	N/A	N/A
10966939	12/2020	Guy et al.	N/A	N/A
11000486	12/2020	Liu et al.	N/A	N/A
11065209	12/2020	Guy et al.	N/A	N/A
11065227	12/2020	Stott et al.	N/A	N/A
11096905	12/2020	Guy et al.	N/A	N/A
11147776	12/2020	Stott et al.	N/A	N/A
11147783	12/2020	Stott et al.	N/A	N/A
11154516	12/2020	Guy et al.	N/A	N/A
11154517	12/2020	Wright et al.	N/A	N/A
11160757	12/2020	Wilkhu et al.	N/A	N/A
11160795	12/2020	Guy et al.	N/A	N/A
11207292	12/2020	Guy et al.	N/A	N/A
11229612	12/2021	Wright et al.	N/A	N/A
11291631	12/2021	Shah	N/A	N/A
11311498	12/2021	Guy et al.	N/A	N/A
11318109	12/2021	Whalley et al.	N/A	N/A
11357741	12/2021	Guy et al.	N/A	N/A
11400055	12/2021	Guy et al.	N/A	N/A
11406623	12/2021	Guy et al.	N/A	N/A
11413266	12/2021	Biro et al.	N/A	N/A
11419829	12/2021	Whalley et al.	N/A	N/A
11426362	12/2021	Wright et al.	N/A	N/A
11446258	12/2021	Guy et al.	N/A	N/A
11590087	12/2022	Guy et al.	N/A	N/A
11633369	12/2022	Guy et al.	N/A	N/A
11679087	12/2022	Guy et al.	N/A	N/A
11684598	12/2022	Stott et al.	N/A	N/A
11701330	12/2022	Guy et al.	N/A	N/A
11766411	12/2022	Guy et al.	N/A	N/A
11793770	12/2022	Stott et al.	N/A	N/A
11806319	12/2022	Wilkhu et al.	N/A	N/A
11865102	12/2023	Guy et al.	N/A	N/A
11963937	12/2023	Guy et al.	N/A	N/A

12023305	12/2023	Whalley et al.	N/A	N/A
12064398	12/2023	Wright et al.	N/A	N/A
12064399	12/2023	Guy et al.	N/A	N/A
2015/0274623	12/2014	Makriyannis et al.	N/A	N/A
2015/0359756	12/2014	Guy et al.	N/A	N/A
2017/0008868	12/2016	Dialer et al.	N/A	N/A
2017/0239193	12/2016	Guy et al.	N/A	N/A
2018/0071210	12/2017	Wilkhu et al.	N/A	N/A
2018/0228751	12/2017	Stott et al.	N/A	N/A
2019/0167583	12/2018	Shah	N/A	N/A
2019/0314296	12/2018	Wright et al.	N/A	N/A
2019/0321307	12/2018	Guy et al.	N/A	N/A
2019/0365667	12/2018	Wright et al.	N/A	N/A
2020/0138738	12/2019	Guy et al.	N/A	N/A
2020/0179303	12/2019	Guy et al.	N/A	N/A
2020/0206153	12/2019	Whalley et al.	N/A	N/A
2020/0237683	12/2019	Whalley et al.	N/A	N/A
2020/0297656	12/2019	Guy et al.	N/A	N/A
2020/0352878	12/2019	Guy et al.	N/A	N/A
2021/0015789	12/2020	Guy et al.	N/A	N/A
2021/0052512	12/2020	Guy et al.	N/A	N/A
2021/0059949	12/2020	Wilkhu et al.	N/A	N/A
2021/0059960	12/2020	Wilkhu et al.	N/A	N/A
2021/0059976	12/2020	Wilkhu et al.	N/A	N/A
2021/0069333	12/2020	Velasco Diez et al.	N/A	N/A
2021/0169824	12/2020	Guy et al.	N/A	N/A
2021/0177773	12/2020	Guy et al.	N/A	N/A
2021/0290565	12/2020	Guy et al.	N/A	N/A
2021/0330636	12/2020	Guy et al.	N/A	N/A
2021/0401771	12/2020	Guy et al.	N/A	N/A
2022/0000800	12/2021	Guy et al.	N/A	N/A
2022/0008355	12/2021	Guy et al.	N/A	N/A
2022/0016048	12/2021	Guy et al.	N/A	N/A
2022/0023232	12/2021	Guy et al.	N/A	N/A
2022/0040155	12/2021	Guy et al.	N/A	N/A
2022/0062197	12/2021	Stott et al.	N/A	N/A
2022/0062211	12/2021	Stott et al.	N/A	N/A
2022/0087951	12/2021	Guy et al.	N/A	N/A
2022/0096397	12/2021	Wright et al.	N/A	N/A
2022/0168266	12/2021	Guy et al.	N/A	N/A
2022/0183997	12/2021	Guy et al.	N/A	N/A
2022/0184000	12/2021	Guy et al.	N/A	N/A
2022/0202738	12/2021	Guy et al.	N/A	N/A
2022/0211629	12/2021	Wilkhu et al.	N/A	N/A
2022/0226257	12/2021	Guy et al.	N/A	N/A
2022/0233495	12/2021	Silcock et al.	N/A	N/A
2022/0249396	12/2021	Guy et al.	N/A	N/A
2022/0257529	12/2021	Guy et al.	N/A	N/A
2022/0265573	12/2021	Guy et al.	N/A	N/A
2022/0288055	12/2021	Silcock et al.	N/A	N/A

2022/0362149	12/2021	Shah	N/A	N/A
2022/0378714	12/2021	Guy et al.	N/A	N/A
2022/0378715	12/2021	Guy et al.	N/A	N/A
2022/0378738	12/2021	Guy et al.	N/A	N/A
2022/0387347	12/2021	Whalley et al.	N/A	N/A
2022/0395470	12/2021	Whalley et al.	N/A	N/A
2022/0395471	12/2021	Guy et al.	N/A	N/A
2023/0000789	12/2022	Guy et al.	N/A	N/A
2023/0022487	12/2022	Guy et al.	N/A	N/A
2023/0024312	12/2022	Whalley et al.	N/A	N/A
2023/0026079	12/2022	Guy et al.	N/A	N/A
2023/0032502	12/2022	Guy et al.	N/A	N/A
2023/0038423	12/2022	Silcock et al.	N/A	N/A
2023/0068885	12/2022	Guy et al.	N/A	N/A
2023/0143812	12/2022	Knappertz et al.	N/A	N/A
2023/0235825	12/2022	Thompson et al.	N/A	N/A
2023/0248664	12/2022	Guy et al.	N/A	N/A
2023/0263744	12/2022	Guy et al.	N/A	N/A
2023/0277560	12/2022	Checketts et al.	N/A	N/A
2023/0277561	12/2022	Checketts et al.	N/A	N/A
2023/0277562	12/2022	Checketts et al.	N/A	N/A
2023/0277563	12/2022	Checketts et al.	N/A	N/A
2023/0285419	12/2022	Checketts et al.	N/A	N/A
2023/0285420	12/2022	Checketts et al.	N/A	N/A
2023/0285421	12/2022	Checketts et al.	N/A	N/A
2023/0285422	12/2022	Checketts et al.	N/A	N/A
2023/0285423	12/2022	Checketts et al.	N/A	N/A
2023/0285424	12/2022	Checketts et al.	N/A	N/A
2023/0285425	12/2022	Checketts et al.	N/A	N/A
2023/0285426	12/2022	Checketts et al.	N/A	N/A
2023/0285427	12/2022	Checketts et al.	N/A	N/A
2023/0285428	12/2022	Checketts et al.	N/A	N/A
2023/0301934	12/2022	Whalley et al.	N/A	N/A
2023/0301936	12/2022	Guy et al.	N/A	N/A
2023/0310464	12/2022	Checketts et al.	N/A	N/A
2023/0372367	12/2022	Checketts et al.	N/A	N/A
2023/0372368	12/2022	Checketts et al.	N/A	N/A
2024/0016819	12/2023	Craig et al.	N/A	N/A
2024/0025858	12/2023	Silcock et al.	N/A	N/A
2024/0033229	12/2023	Guy et al.	N/A	N/A
2024/0033272	12/2023	Checketts et al.	N/A	N/A
2024/0043388	12/2023	Silcock et al.	N/A	N/A
2024/0050452	12/2023	Craig et al.	N/A	N/A
2024/0091241	12/2023	Guy et al.	N/A	N/A
2024/0130981	12/2023	Wilkhu et al.	N/A	N/A
2024/0131041	12/2023	Tse et al.	N/A	N/A
2024/0165048	12/2023	Guy et al.	N/A	N/A
2024/0207220	12/2023	Guy et al.	N/A	N/A
2024/0215910	12/2023	Tse et al.	N/A	N/A
2024/0226032	12/2023	Wilkhu et al.	N/A	N/A

2024/0226123	12/2023	Tse et al.	N/A	N/A
2024/0238218	12/2023	Silcock et al.	N/A	N/A
2024/0254066	12/2023	Silcock et al.	N/A	N/A
2024/0254072	12/2023	Silcock et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3 061 450	12/2015	EP	N/A
WO 01/95899	12/2000	WO	N/A
WO 2014/062965	12/2013	WO	N/A
WO 2015/193667	12/2014	WO	N/A
WO 2015/193668	12/2014	WO	N/A
WO 2016/094810	12/2015	WO	N/A
WO 2016/135308	12/2015	WO	N/A
WO 2016/203239	12/2015	WO	N/A
WO 2018/061007	12/2017	WO	N/A
WO 2018/205022	12/2017	WO	N/A

OTHER PUBLICATIONS

Namba K, Yamamoto H, Sasaki I, Mori K, Imagawa H, Nishizawa M. Hg (OTf) 2-catalyzed arylene cyclization. *Organic Letters*. May 1, 2008;10(9):1767-70. (Year: 2008). cited by examiner

Carral-Menoyo A, Misol A, Gómez-Redondo M, Sotomayor N, Lete E. Palladium (II)-Catalyzed Intramolecular C—H Alkenylation for the Synthesis of Chromanes. *The Journal of Organic Chemistry*. Jan. 14, 2019;84(4):2048-60. (Year: 2019). cited by examiner

Schafroth MA, Rummelt SM, Sarlah D, Carreira EM. Enantioselective Iridium-catalyzed allylic cyclizations. *Organic letters*. Jun. 16, 2017;19(12):3235-8. (Year: 2017). cited by examiner

Huffman JW, Yu S. Synthesis of a tetracyclic, conformationally constrained analogue of Δ^8 -THC. *Bioorganic & medicinal chemistry*. Dec. 1, 1998;6(12):2281-8. (Year: 1998). cited by examiner

Friedman D, Devinsky O. Cannabinoids in the treatment of epilepsy. *New England Journal of Medicine*. Sep. 10, 2015;373(11):1048-58. (Year: 2015). cited by examiner

Kimball, A. W. et al., "Chemical Protection against Ionizing Radiation," *Radiation Research*, 7:1-12 (1957). cited by applicant

Litchfield, J. T. & Wilcoxon, F., "A simplified method of evaluating dose-effect experiments," *Fed Proc*. Mar. 1948; 7(1 Pt 1):240, 15 pages. cited by applicant

Loscher, W. et al., "The role of technical, biological and pharmacological factors in the laboratory evaluation of anticonvulsant drugs. II. Maximal electroshock seizure models," *Epilepsy Res*. Mar. 1991; 8(2):79-94. doi: 10.1016/0920-1211(91)90075-q. cited by applicant

Epidiolex® (cannabidiol) oral solution, CV, Prescribing Information, 2018, 30 pages; https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/210365lbl.pdf. cited by applicant

Harvey, D. J., "Characterization of the butyl homologues of Δ^1 -tetrahydrocannabinol, cannabinol and cannabidiol in samples of cannabis by combined gas chromatography and mass spectrometry," *J. Pharm. Pharmac.*, 28:280-285 (1976). cited by applicant

Hill, T. D. M. et al., "Cannabidiol-rich cannabis extracts are anticonvulsant in mouse and rat via a CB1 receptor-independent mechanism," *British Journal of Pharmacology*, 170:679-692 (2013). cited by applicant

Morales, P. et al., "An Overview on Medicinal Chemistry of Synthetic and Natural Derivatives of Cannabidiol," *Frontiers in Pharmacology*, 8:422 (2017); doi:10.3389/fphar.2017.00422, 18 pages. cited by applicant

Pertwee, "The Pharmacology and Therapeutic Potential of Cannabidiol," *Cannabinoids*, Chapter 3, DiMarzo, V. (Ed.), pp. 32-83 (2004). cited by applicant

Vree et al., "Identification of hashish of tetrahydrocannabinol, cannabidiol and cannabinol analogues with a methyl side-chain," J. Pharm. Pharmac. 24:7-12 (1972). cited by applicant
U.S. Appl. No. 15/640,033, filed Jun. 30, 2017. cited by applicant
U.S. Appl. No. 16/467,639, filed Jun. 7, 2019. cited by applicant
U.S. Appl. No. 16/768,241, filed May 29, 2020. cited by applicant
U.S. Appl. No. 16/959,350, filed Jun. 30, 2020. cited by applicant
U.S. Appl. No. 16/959,354, filed Jun. 30, 2020. cited by applicant
U.S. Appl. No. 16/959,357, filed Jun. 30, 2020. cited by applicant
U.S. Appl. No. 17/050,956, filed Oct. 27, 2020. cited by applicant
U.S. Appl. No. 16/935,005, filed Jul. 21, 2020. cited by applicant
U.S. Appl. No. 17/296,066, filed May 21, 2021. cited by applicant
U.S. Appl. No. 17/296,076, filed May 21, 2021. cited by applicant
U.S. Appl. No. 17/424,682, filed Jul. 21, 2021. cited by applicant
U.S. Appl. No. 17/426,442, filed Jul. 28, 2021. cited by applicant
U.S. Appl. No. 17/406,401, filed Aug. 19, 2021. cited by applicant
U.S. Appl. No. 17/435,892, filed Sep. 2, 2021. cited by applicant
U.S. Appl. No. 17/470,382, filed Sep. 9, 2021. cited by applicant
U.S. Appl. No. 17/472,000, filed Sep. 10, 2021. cited by applicant
U.S. Appl. No. 17/472,016, filed Sep. 10, 2021. cited by applicant
U.S. Appl. No. 17/548,232, filed Dec. 10, 2021. cited by applicant
U.S. Appl. No. 17/606,370, filed Oct. 25, 2021. cited by applicant
U.S. Appl. No. 17/611,824, filed Nov. 16, 2021. cited by applicant
U.S. Appl. No. 17/529,005, filed Nov. 17, 2021. cited by applicant
U.S. Appl. No. 17/615,422, filed Nov. 30, 2021. cited by applicant
U.S. Appl. No. 17/552,487, filed Dec. 16, 2021. cited by applicant
U.S. Appl. No. 17/631,069, filed Jan. 28, 2022. cited by applicant
U.S. Appl. No. 17/638,629, filed Feb. 25, 2022. cited by applicant
U.S. Appl. No. 17/689,607, filed Mar. 8, 2022. cited by applicant
U.S. Appl. No. 17/689,245, filed Mar. 8, 2022. cited by applicant
U.S. Appl. No. 17/744,224, filed May 13, 2022. cited by applicant
U.S. Appl. No. 17/705,443, filed Mar. 28, 2022. cited by applicant
U.S. Appl. No. 17/680,048, filed Apr. 11, 2022. cited by applicant
U.S. Appl. No. 17/770,435, filed Apr. 20, 2022. cited by applicant
U.S. Appl. No. 17/770,436, filed Apr. 20, 2022. cited by applicant
U.S. Appl. No. 17/771,184, filed Apr. 22, 2022. cited by applicant
U.S. Appl. No. 17/771,190, filed Apr. 22, 2022. cited by applicant
U.S. Appl. No. 17/771,195, filed Apr. 22, 2022. cited by applicant
U.S. Appl. No. 17/771,183, filed Apr. 22, 2022. cited by applicant
U.S. Appl. No. 17/777,734, filed May 18, 2022. cited by applicant
U.S. Appl. No. 17/777,677, filed May 18, 2022. cited by applicant
U.S. Appl. No. 17/777,681, filed May 18, 2022. cited by applicant
U.S. Appl. No. 17/786,949, filed Jun. 17, 2022. cited by applicant
U.S. Appl. No. 17/853,367, filed Jun. 29, 2022. cited by applicant
U.S. Appl. No. 17/816,349, filed Jul. 29, 2022. cited by applicant
U.S. Appl. No. 17/819,046, filed Aug. 11, 2022. cited by applicant
U.S. Appl. No. 18/005,838, filed Jan. 17, 2023. cited by applicant
U.S. Appl. No. 18/005,841, filed Jan. 17, 2023. cited by applicant
U.S. Appl. No. 18/005,845, filed Jan. 17, 2023. cited by applicant
U.S. Appl. No. 18/005,843, filed Jan. 17, 2023. cited by applicant
U.S. Appl. No. 18/005,847, filed Jan. 17, 2023. cited by applicant

U.S. Appl. No. 18/005,848, filed Jan. 17, 2023. cited by applicant
U.S. Appl. No. 18/005,851, filed Jan. 18, 2023. cited by applicant
U.S. Appl. No. 18/005,852, filed Jan. 18, 2023. cited by applicant
U.S. Appl. No. 18/005,853, filed Jan. 18, 2023. cited by applicant
U.S. Appl. No. 18/005,959, filed Jan. 18, 2023. cited by applicant
U.S. Appl. No. 18/005,960, filed Jan. 18, 2023. cited by applicant
U.S. Appl. No. 18/005,961, filed Jan. 18, 2023. cited by applicant
U.S. Appl. No. 18/006,125, filed Jan. 19, 2023. cited by applicant
U.S. Appl. No. 18/006,127, filed Jan. 19, 2023. cited by applicant
U.S. Appl. No. 18/006,129, filed Jan. 19, 2023. cited by applicant
U.S. Appl. No. 18/006,131, filed Jan. 19, 2023. cited by applicant
U.S. Appl. No. 18/006,133, filed Jan. 19, 2023. cited by applicant
U.S. Appl. No. 18/006,121, filed Jan. 19, 2023. cited by applicant
U.S. Appl. No. 18/161,603, filed Jan. 30, 2023. cited by applicant
U.S. Appl. No. 18/170,235, filed Feb. 16, 2023. cited by applicant
U.S. Appl. No. 18/043,810, filed Mar. 2, 2023. cited by applicant
U.S. Appl. No. 18/044,941, filed Mar. 10, 2023. cited by applicant
U.S. Appl. No. 18/245,856, filed Mar. 17, 2023. cited by applicant
U.S. Appl. No. 18/186,792, filed Mar. 20, 2023. cited by applicant

Primary Examiner: Lundgren; Jeffrey S

Assistant Examiner: Dahlin; Heather

Attorney, Agent or Firm: COOLEY LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of International PCT Application No. PCT/GB2020/051723, filed Jul. 17, 2020; and Great Britain Application No. 1910389.4, filed Jul. 19, 2019; each of the aforementioned applications are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(2) The present invention relates to a group of cannabinoid derivatives as pharmaceutically active compounds and methods of preparation thereof.

(3) The cannabinoid derivatives of the invention are analogues of cannabidiol (CBD). CBD is a non-psychoactive cannabinoid which has been used to treat various diseases and disorders. While such treatments hold promise, there remains a need in the art for more effective treatments and this has been brought about by way of the cannabinoid derivatives of the invention.

BACKGROUND TO THE INVENTION

(4) Cannabinoids are natural and synthetic compounds structurally or pharmacologically related to the constituents of the cannabis plant or to the endogenous agonists (endocannabinoids) of the cannabinoid receptors CB1 or CB2. The only way in nature in which these compounds are produced is by the cannabis plant. Cannabis is a genus of flowering plants in the family *Cannabaceae*, comprising the species *Cannabis sativa*, *Cannabis indica*, and *Cannabis ruderalis* (sometimes considered as part of *Cannabis sativa*).

(5) Cannabis plants comprise a highly complex mixture of compounds. At least 568 unique molecules have been identified. Among these compounds are cannabinoids, terpenoids, sugars,

fatty acids, flavonoids, other hydrocarbons, nitrogenous compounds, and amino acids.

(6) Cannabinoids exert their physiological effects through a variety of receptors including, but not limited to, adrenergic receptors, cannabinoid receptors (CB1 and CB2), GPR55, GPR3, or GPR8. The principle cannabinoids present in cannabis plants are cannabinoid acids Δ^9 -tetrahydrocannabinolic acid (Δ^9 -THCA) and cannabidiolic acid (CBDA) with small amounts of their respective neutral (decarboxylated) cannabinoids. In addition, cannabis may contain lower levels of other minor cannabinoids.

(7) There are currently four cannabinoid-based pharmaceutical approved products on the market. These are: dronabinol (Marinol®) which is a synthetic tetrahydrocannabinol (THC) approved for the treatment of loss of appetite in AIDS and the treatment of severe nausea and vomiting caused by cancer chemotherapy; nabilone (Cesamet®) which is a synthetic cannabinoid and an analog of THC which is approved for the treatment of nausea and vomiting caused by cytotoxic chemotherapy unresponsive to conventional antiemetics; nabiximols (Sativex®) a mixture of two cannabis plant extracts approved for the treatment of neuropathic pain, spasticity, overactive bladder, and other symptoms of multiple sclerosis; and highly purified botanical CBD (Epidiolex®) approved in the United States for the treatment of Dravet syndrome and Lennox-Gastaut syndrome in children and adults over the age of 2 years.

(8) As can be seen above cannabinoids are a class of compounds which may be derived naturally from the cannabis plant or produced semi-synthetically or synthetically via chemical synthesis.

(9) More than 100 different cannabinoids have been identified. These cannabinoids can be split into different groups as follows: phytocannabinoids; endocannabinoids and synthetic cannabinoids (which may be novel cannabinoids or synthetically produced versions of phytocannabinoids or endocannabinoids). The Handbook of Cannabis, Roger Pertwee, Chapter 1, pages 3 to 15 details the cannabinoids known to date.

(10) Cannabidiol (CBD) is a major cannabinoid constituent of Cannabis species, such as the hemp plant (*Cannabis sativa*). Unlike other cannabinoids, such as THC, cannabidiol does not bind to CB1 or CB2 receptors, or its binding to the receptors is negligible in terms of inducing a pharmacological effect. Thus, cannabidiol does not cause the central or peripheral nervous system effects mediated by the CB1 or CB2 receptors. CBD has little or no psychotropic (cannabimimetic) activity and its molecular structure and properties are substantially different from those of other cannabinoids.

(11) Cannabidiol administration has been the subject of research in an attempt to provide an alternative treatment for various diseases and disorders which may respond to such treatment.

(12) The synthetic production of the metabolite of CBD, 7-hydroxy-cannabidiol, (7-OH CBD) is disclosed in WO 01/95899. The compound was tested in a model of inflammation and found to be effective. The application goes on to suggest that the compound may be of use as an analgesic, anti-anxiety, anti-convulsant, neuroprotective, anti-psychotic and anti-inflammatory based on the mechanisms the compound displays in the model of inflammation.

(13) The present invention relates to novel cannabinoid compounds which are biologically active and hence useful in the treatment of diseases. Such novel compounds may be administered by a wide variety of routes including but not limited to oral, transdermal, buccal, nasal, pulmonary, rectal or ocular. Such compounds may be used for the treatment or prevention of a medical condition such as epilepsy, pain, inflammation and cancer.

BRIEF SUMMARY OF THE DISCLOSURE

(14) In accordance with a first aspect of the present invention there is provided a compound of general Formula I or a salt thereof,

(15) ##STR00001##

where X is either CH₂; O; NBoc; NFMoc; NZ; NTs; NAc; NC(O) iPr; NBz or NH and Y is either H or OH.

(16) In accordance with a second aspect of the present invention there is provided a compound of

general Formula II or a salt thereof,

(17) ##STR00002##

where X is either CH₂; O; NBoc; NFMoc; NZ; NTs; NAc; NC(O)*i*Pr; NBz or NH and Y is either H or OH.

(18) In accordance with a third aspect of the present invention there is provided a pharmaceutical composition comprising the compound of general Formula I or Formula II.

(19) Preferably the pharmaceutical composition is selected from a tablet, a capsule, a granule, an oral solution, a powder for inhalation, a sprinkle, an oral solution and a suspension.

(20) Preferably the pharmaceutical composition comprises one or more of: an excipient selected among a carrier, an oil, a disintegrant, a lubricant, a stabilizer, a flavouring agent, an antioxidant, a diluent and another pharmaceutically effective compound.

(21) In accordance with a fourth aspect of the present invention there is provided a compound of general Formula I or Formula II for use as a medicament.

(22) In accordance with a fifth aspect of the present invention there is provided a compound of general Formula I or Formula II for use in the treatment of epilepsy.

(23) In accordance with a sixth aspect of the present invention there is provided a comprising administering a pharmaceutical preparation comprising a compound of general Formula I or Formula II.

(24) In accordance with a seventh aspect of the present invention there is provided a process for the production of a compound of general Formula I or Formula II comprising reacting a resorcinol unit of Structure 1a-j via a Friedel-Crafts 1,4-addition to produce compounds of Structures 2a to 2j or 3a to 3j followed by subsequent steps to produce the compounds of general Formula I or Formula II via intermediates.

(25) In accordance with an eighth aspect of the present invention there is provided an intermediate formed in the process of the production of a compound of general Formula I or Formula II.

DEFINITIONS

(26) "Cannabinoids" are a group of compounds including the endocannabinoids, the phytocannabinoids and those which are neither endocannabinoids or phytocannabinoids, hereinafter "syntho-cannabinoids".

(27) "Endocannabinoids" are endogenous cannabinoids, which are high affinity ligands of CB1 and CB2 receptors.

(28) "Phytocannabinoids" are cannabinoids that originate in nature and can be found in the cannabis plant. The phytocannabinoids can be present in an extract including a botanical drug substance, isolated, or reproduced synthetically.

(29) "Syntho-cannabinoids" are those compounds that are not found endogenously or in the cannabis plant. Examples include WIN 55212 and rimonabant.

(30) An "isolated phytocannabinoid" is one which has been extracted from the cannabis plant and purified to such an extent that all the additional components such as secondary and minor cannabinoids and the non-cannabinoid fraction have been removed.

(31) A "synthetic cannabinoid" is one which has been produced by chemical synthesis. This term includes modifying an isolated phytocannabinoid, by, for example, forming a pharmaceutically acceptable salt thereof.

(32) A "substantially pure" cannabinoid is defined as a cannabinoid which is present at greater than 95% (w/w) pure. More preferably greater than 96% (w/w) through 97% (w/w) thorough 98% (w/w) to 99% (w/w) and greater.

DETAILED DESCRIPTION OF THE INVENTION

(33) The following describes the production of the novel cannabinoid derivatives as claimed in the present invention. The structure and absolute configuration of the compounds produced were determined by X-ray crystallographic analysis, using a molybdenum X-Ray source.

Description

Example 1: Method of Manufacture of Normal CBD Derivatives

(1) This example describes a novel method of synthesis which was used to produce novel analogues of normal CBD which demonstrated pharmacological activity. Scheme 1 below describes the initial reaction which was used to produce the primary intermediate and Scheme 2 describes the production of the normal CBD derivatives which were formed via a number of intermediates.

(2) ##STR00003##

Where R.sub.1=H or OMe and X=CH.sub.2; O; NBoc; NFMoc; NZ; NTs; NAc; NC(O)iPr; NBz or NH

(3) The resorcinol unit, in Structures 1a to 1j (shown below), underwent a Friedel-Crafts 1,4-addition reaction as shown in Scheme 1 to produce compounds of Structures 2a to 2j described below.

(4) Various catalysts were tested. Silylated Jorgensen-Hayashi type gave poor yields but excellent selectivity. MacMillan-type catalysts gave better yields but poorer enantioselectivities.

(5) ##STR00004## ##STR00005## ##STR00006## ##STR00007## ##STR00008##

(6) The compound of Structures 2a to 2j or 3a to 3j (shown above) were then reacted as shown in Scheme 2 below to derive two different normal CBD derivatives 11a to 11j or 12a to 12j.

(7) Deprotection of the derivatives 11c-j and 12 c-j further produced compounds 13 and 14.

(8) ##STR00009##

Synthesis of Intermediates 4a to 4j or 5a to 5j

(9) Intermediates 2a-2j or 3a-3j were brominated with N-bromosuccinimide (NBS) to create arylbromide system compounds 4a-4j or 5a-5j shown below.

(10) ##STR00010## ##STR00011## ##STR00012## ##STR00013##

Synthesis of Intermediates 6a to 6j or 7a to 7j

(11) Intermediates 4a-4j or 5a-5j were O-demethylated with boron tribromide in the presence of dichloromethane which resulted in the deprotected chiral resorcinol compounds as described in 6a to 6j or 7a to 7j depicted below.

(12) ##STR00014## ##STR00015## ##STR00016## ##STR00017##

Synthesis of Intermediates 9a to 9j (except 9f) or 10a to 10j (except 10f)

(13) Coupling of the brominated intermediates of 6a-6j or 7a-7j with menthadienol (shown as structure 8 in Scheme 2) in the presence of boron trifluoride diethyl etherate produced the intermediates 9a to 9j (except 9f which is not possible) or 10a to 10j (except 10f which is not possible). These structures are depicted below.

(14) ##STR00018## ##STR00019## ##STR00020## ##STR00021##

Synthesis of Normal CBD Derivatives 11a to 11j (Except 11f) or 12a to 12j (except 12f)

(15) 9a to 9j (except 9f) or 10a to 10j (except 10f) were debrominated with acetic acid and hydrogen bromide to produce the normal CBD derivatives 11a to 11j (except 11f) or 12a to 12j (except 12f) shown below.

(16) ##STR00022## ##STR00023## ##STR00024##

(17) Furthermore, the N-protected moieties compounds 11c-11j (except 11f) and 12c to 12j (except 12f) can be deprotected to give the two NH derivatives, compounds 13 and 14, shown below.

(18) ##STR00025##

Example 2: Method of Manufacture of Abnormal CBD Derivatives

(19) This example describes a novel method of synthesis which was used to produce novel analogues of abnormal CBD which demonstrated pharmacological activity. Scheme 1 depicted in Example 1 describes the initial reaction which was used to produce the primary intermediates 2a to 2j and 3a to 3j and Scheme 3 describes the production of the abnormal CBD derivatives which

were formed via a number of intermediates.

(20) ##STR00026##

Synthesis of Intermediates 15a to 15j or 16a to 16j

(21) Intermediates 2a to 2j or 3a to 3j were O-demethylated with boron tribromide in the presence of dichloromethane which resulted in the deprotected chiral resorcinol compounds as described in 15a to 15j or 16a to 16j below.

(22) ##STR00027## ##STR00028## ##STR00029##

Synthesis of Abnormal CBD Derivatives 17a to 17j or 18a to 18j

(23) Coupling of the intermediates of 15a-15j or 16a-16j with menthadienol (shown as structure 8 in Scheme 3) in the presence of boron trifluoride diethyl etherate produced the abnormal CBD derivatives 17a to 17j or 18a to 18j depicted below.

(24) ##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034##
##STR00035##

Synthesis of Compounds 19 to 22

(25) Deprotection of 17c to 17j or 18c to 18j produced the compounds 19 to 22 described below.

(26) ##STR00036##

Conclusion

(27) The application of the novel routes of synthesis to produce novel cannabidiol analogues and their intermediates is of benefit.

(28) Compounds of the general formulas I and II are as detailed below and are equivalent to the compounds of structures 11a to 11j (except 11f), 12a to 12j (except 11f), 17a to 17j and 17a to 17j. Such compounds may provide improved or new therapeutic treatment options.

(29) Deprotection of particular compounds has been found to produce additional novel molecules which provide additional improved or new therapeutic benefit. Such compounds include 13, 14, 19, 20, 21 and 22.

(30) ##STR00037##

where X is either CH₂; O; NBoc; NFMoc; NZ; NTs; NAc; NC(O)*i*Pr; NBz or NH and Y is either H or OH.

(31) ##STR00038##

where X is either CH₂; O; NBoc; NFMoc; NZ; NTs; NAc; NC(O)*i*Pr; NBz or NH and Y is either H or OH.

Example 3: Evaluation of Cannabinoid Derivatives for Anticonvulsant Activity using the Maximal Electroshock Seizure Threshold (MEST) Test in the Mouse

(32) The efficacy of exemplary cannabinoid derivatives according to Formula I and Formula II were tested in a mouse model of generalised seizure, the maximal electroshock seizure threshold (MEST) test.

(33) The maximal electroshock seizure threshold (MEST) test is widely utilized preclinically to evaluate pro- or anti-convulsant properties of test compounds (Loscher et al., 1991).

(34) In the MEST test the ability of a drug to alter the seizure threshold current required to induce hind limb tonic extensor convulsions is measured according to an "up and down" method of shock titration (Kimball et al., 1957). An increase in seizure threshold is indicative of anti-convulsant effect. Antiepileptic drugs including the sodium channel blockers (e.g. lamotrigine) with clinically proven efficacy against generalised tonic-clonic seizures all exhibit anti-convulsant properties in this test in the mouse.

(35) Conversely, a reduction in seizure threshold is indicative of a pro-convulsant effect as observed with known convulsant agents such as picrotoxin.

(36) The ability of a test compound to alter the stimulus intensity, expressed as current (mA), required to induce the presence of tonic hind limb extensor convulsions, is assessed in the MEST. The outcome of the presence (+) or absence (0) of tonic hind limb extensor convulsions observed from a current to produce tonic hind limb extension in 50% of animals in the treatment group

(CC.sub.50) determines the seizure threshold for the treatment group and the effects were then compared to the CC.sub.50 of the vehicle control group.

Methods

Study Details

(37) Naïve mice were acclimatised to the procedure room in their home cages for up to 7 days, with food and water available ad libitum.

(38) All animals were weighed at the beginning of the study and randomly assigned to treatment groups based on a mean distribution of body weight across groups. All animals were dosed at 10 mL/kg via intraperitoneal (i.p) injection, with either vehicle, test compound at 200 mg/kg, or diazepam at 2.5 mg/kg.

(39) Animals were individually assessed for the production of a tonic hind limb extensor convulsion at 60 min post-dose for vehicle, 30-120 min post-dose for test compound (dependant on compound) and 30 min post-dose for diazepam, from a single electroshock.

(40) The first animal within a treatment group was given a shock at the expected or estimated CC.sub.50 current. For subsequent animals, the current was lowered or raised depending on the convulsions outcome from the preceding animal.

(41) Data generated from each treatment group were used to calculate the CC.sub.50±SEM values for the treatment group.

Test Compounds

(42) Vehicle: (5% ethanol, 5% solutol in 90% Saline) was prepared as follows: 2 mL of ethanol, 2 mL of solutol were warmed to 60° C., in 36 mL of saline (1:1:18).

(43) Positive control: diazepam was used at 2.5 mg/kg.

(44) The test compounds used were 12a, 12b, 18a and 18b. Test compounds were administered at 200 mg/kg (i.p.) in a 1:1:18 ethanol:solutol:saline formulation.

Sample Collection

(45) Each animal was humanely killed immediately after production of a convulsion by destruction of the brain from striking the cranium, followed by the confirmation of permanent cessation of the circulation from decapitation under The Humane Killing of Animals under Schedule 1 to the Animals (Scientific Procedures) Act 1986. Terminal blood and brain collection were performed following decapitation.

(46) Blood was collected in Lithium-heparin tubes and centrifuged at 4° C. for 10 minutes at 1500×g. The resulting plasma was removed (>100 µL) and split into 2 aliquots of 0.5 mL Eppendorf tubes containing 10 µL of ascorbic acid (100 mg/mL) for stabilisation. Brains were removed, washed in saline and halved. Each half was placed into separate 2 mL screw cap cryovials, weighed and frozen on cardice.

Statistical analysis

(47) The data for each treatment group were recorded as the number of +'s and 0's at each current level employed and this information is then used to calculate the CC.sub.50 value (current required for 50% of the animals to show seizure behaviour)±standard error.

(48) Test compound effects were also calculated as percentage change in CC.sub.50 from the vehicle control group.

(49) Significant difference between drug-treated animals and controls were assessed according to Litchfield and Wilcoxon (1949).

Results

(50) FIGS. 1 to 4 and Tables 1 to 4 describe the data produced in this experiment.

(51) In the vehicle group, the CC.sub.50 value was calculated to be 21 mA.

(52) In the diazepam (2.5 mg/kg) treated group, administered i.p. 30 minutes before the test, the CC.sub.50 value was 35 mA. This result was statistically significant ($p<0.001$) compared to the vehicle control.

(53) In the test compound treatment groups, administered i.p. between 30 and 120 minutes before

the test, all four compounds produced a statistically significant CC.sub.50 value compared to vehicle.

(54) Such data are indicative that these compounds will be of therapeutic benefit.

(55) TABLE-US-00001 TABLE 1 Evaluation of effect of Compound 12a in the MEST test
Test time % change Dose post dose CC.sub.50 +/- Sig- from Treatment (mg/kg) (min) N SEM nificance
vehicle Vehicle — 60 12 21.0 +/- 0.5 — — Diazepam 2.5 30 12 35.0 +/- 1.1 P < 0.001 66%
Compound 200 60 8 54.0 +/- 0.2 P < 0.001 157% 12a

(56) TABLE-US-00002 TABLE 2 Evaluation of effect of Compound 12b in the MEST test
Test time % change Dose post dose CC.sub.50 +/- Sig- from Treatment (mg/kg) (min) N SEM nificance
vehicle Vehicle — 60 12 21.0 +/- 0.5 — — Diazepam 2.5 30 12 35.0 +/- 1.1 P < 0.001 66%
Compound 200 30 12 43.8 +/- 0.2 P < 0.001 109% 12b

(57) TABLE-US-00003 TABLE 3 Evaluation of effect of Compound 18a in the MEST test
Test time % change Dose post dose CC.sub.50 +/- Sig- from Treatment (mg/kg) (min) N SEM nificance
vehicle Vehicle — 60 12 21.0 +/- 0.5 — — Diazepam 2.5 30 12 35.0 +/- 1.1 P < 0.001 66%
Compound 200 120 12 41.8 +/- 0.5 P < 0.001 99% 18a

(58) TABLE-US-00004 TABLE 4 Evaluation of effect of Compound 18b in the MEST test
Test time % change Dose post dose CC.sub.50 +/- Sig- from Treatment (mg/kg) (min) N SEM nificance
vehicle Vehicle — 60 12 21.0 +/- 0.5 — — Diazepam 2.5 30 12 35.0 +/- 1.1 P < 0.001 66%
Compound 200 120 12 40.3 +/- 0.8 P < 0.001 92% 18b

Conclusions

(59) These data demonstrate a therapeutic effect for the compounds of Formula I and Formula II.

(60) These data are significant as they provide heretofore unknown evidence that these novel cannabinoid derivatives may be of therapeutic value.

(61) The compounds tested were those detailed as Compound 12a, Compound 12b, Compound 18a and Compound 18b. Such compounds are examples of the cannabinoid derivatives of general Formula I and Formula II.

(62) Clearly as all compounds showed efficacy in the MEST test such therapeutic efficacy can be attributed to the cannabinoid derivatives of general Formula I and Formula II of the invention.

Claims

1. A compound of the general Formula I or a salt thereof ##STR00039## where X is either CH.sub.2; O; NBoc; NFMoc; NZ; NTs; NAc; NC (O)iPr; NBz or NH and Y is either H or OH.
2. A compound of general Formula II or a salt thereof ##STR00040## where X is either CH.sub.2; O; NBoc; NFMoc; NZ; NTs; NAc; NC (O) iPr; NBz or NH and Y is either H or OH.
3. A pharmaceutical composition comprising a compound of claim 1 or salt thereof.
4. A pharmaceutical composition as claimed in claim 3, wherein the pharmaceutical composition is selected from a tablet, a capsule, a granule, a powder for inhalation, a sprinkle, an oral solution and a suspension.
5. A pharmaceutical composition as claimed in claim 3, wherein the composition additionally comprises one or more of: an excipient selected among a carrier, an oil, a disintegrant, a lubricant, a stabilizer, a flavouring agent, an antioxidant, a diluent and another pharmaceutically effective compound.
6. A method of treating epilepsy in a mammal in need thereof comprising administering a pharmaceutical preparation comprising a compound of claim 1 or salt thereof.
7. A process for the production of a compound of general Formula I comprising reacting a resorcinol unit of Structure 1a-j via a Friedel-Crafts 1,4-addition to produce compounds of Structures 2a to 2j or 3a to 3j followed by subsequent steps to produce the compounds of general Formula I via intermediates.
8. A compound selected from any one of: ##STR00041## ##STR00042## wherein R.sub.1 is H,

and wherein X is selected from any one of CH.sub.2, O, NBoc, NFmoc, NZ, NTs, NAc, NC(O)iPr and NBz.

9. A pharmaceutical composition comprising a compound of claim 2 or salt thereof.

10. A pharmaceutical composition as claimed in claim 9, wherein the pharmaceutical composition is selected from a tablet, a capsule, a granule, a powder for inhalation, a sprinkle, an oral solution and a suspension.

11. A pharmaceutical composition as claimed in claim 9, wherein the composition additionally comprises one or more of: an excipient selected among a carrier, an oil, a disintegrant, a lubricant, a stabilizer, a flavouring agent, an antioxidant, a diluent and another pharmaceutically effective compound.

12. A method of treating epilepsy in a mammal in need thereof comprising administering a pharmaceutical preparation comprising a compound of claim 2 or a salt thereof.

13. A process for the production of a compound of general Formula II comprising reacting a resorcinol unit of Structure 1a-j via a Friedel-Crafts 1,4-addition to produce compounds of Structures 2a to 2j or 3a to 3j followed by subsequent steps to produce the compounds of general Formula II via intermediates.
